

# Making sense of the “forestry research game” at universities

by David A. MacLean<sup>1</sup>

## ABSTRACT

In this article, I discuss forestry research as conducted in universities. The research process is summarized as a cycle of six steps, including 1) an idea, question, testable hypothesis, or user need; 2) that gets expanded into a proposal and required budget; 3) which is used to seek funding to conduct the project; 4) and also the need to find the right student or personnel, to 5) then do the project, and 6) convey the results. This may lead to more questions and thus restart the “cycle.” Elements that contribute to success at each step of the process and to effective researcher–client partnerships are discussed.

**Key words:** effective forestry research partnerships, research ranking, publications, joint projects between university researchers and industry or government clients

## RÉSUMÉ

Dans cet article, je discute de la recherche forestière telle qu'effectuée dans les universités. Le processus de recherche se résume en un cycle de six étapes, comprenant 1) une idée, une question, une hypothèse vérifiable ou un besoin d'un utilisateur; 2) le tout est traduit en une proposition accompagnée de son budget; 3) qui, à son tour, est utilisée pour chercher des fonds pour réaliser le projet; 4) suivi de l'obligation de trouver le bon étudiant ou le personnel requis afin de, 5) pouvoir entreprendre le projet et, 6) de transmettre les résultats. Le tout peut engendrer de nouvelles questions et ainsi redémarrer le « cycle ». Les éléments qui contribuent au succès de chaque étape du processus et à obtenir une relation efficace entre le client et le chercheur sont discutés.

**Mots clés :** relations efficaces en recherche forestière, classification de la recherche, publications, projets conjoints entre chercheurs universitaires et clients de l'industrie ou du gouvernement



David A. MacLean

## Introduction

I thought about naming this article “*A Dummy’s Guide to Forestry Research at Universities*,” but always thought those titles reflected somewhat badly on the reader, so opted not to. And by using the name “game” in the title, I in no way demean or minimize the importance of forestry research—rather, I see it as having fairly self-evident rules, winners and losers, champions, breakthroughs,

“evangelists” and nay-sayers, etc.

I hope that my thoughts on the topic may be of interest to several groups:

- Forest practitioners, who must sometimes find the whole “research game” rather mystifying
- Graduate students, if it helps to clarify pros and cons of different paths to follow
- Academics wondering why “the rich get richer and the poor get poorer”
- Forest industry and other potential research partners, in terms of the “levers” in the game.

I don’t pretend to have all the answers and others may disagree, but this is my view.

First, a bit about my background: I spent 21 years as a research scientist with the Canadian Forest Service (CFS) before moving to the University of New Brunswick (UNB) as Dean in 1999. In CFS, I purposefully avoided research management, except for a six-month stint as a Program Director, but did “manage” my own research group of four permanent CFS staff and four others funded externally from project “soft money” (i.e., not CFS funds). At UNB as Dean I have maintained a fairly active research program (currently eight graduate students, with nine supervised or co-supervised Masters or PhD students completed from 2004 to 2007). Given Dean responsibilities, much of the research time comes from evenings and weekends, and I guess I do it because I like it. I’m not the only one; several other Canadian forestry school deans or department chairs have active research programs.

Of course, it is impossible to generalize to fit every situation. University research can range from a professor doing some measurements at no cost over a long period of time, to a theoretical or computer analysis, to huge multidisciplinary teams and equipment costing millions of dollars. As frequently noted by Hamish Kimmins about forest ecology, “it depends.” However, typically academic forestry research requires the coinciding of a research idea or need, personnel to work on it (most often a graduate student, sometimes a post-doctoral fellow or other staff or only a professor), and funding to conduct the study and pay for the personnel. Having all three coincide is a major trick. And not just any

<sup>1</sup>Dean, Faculty of Forestry and Environmental Management, University of New Brunswick, P.O. Box 44555, Fredericton, New Brunswick, E3B 5A3. E-mail: macleand@unb.ca

personnel will do—must be the right aptitude, knowledge, skills, and interpersonal relations. Ideally (in my view), this idea–funding–personnel triumvirate may include a fourth component, a partner/client with a research need, who can help with the funding but also has needs and expectations (more about this later).

### The “Forestry Research Game”

I think of the research process as a cycle with six steps (Fig. 1).

1. It starts with an *idea, a question, a testable hypothesis, or a user need*.
2. This gets expanded into a *proposal*, a written description of the problem, what is known about it already, specific methods to be used to address the question, and a budget.
3. This is then “shopped” to a *funder*, perhaps an individual partner/client interested enough in the question to provide funds to do it, or a funding program either specific to the topic area (e.g., a response to a targeted “call for proposals”) or general (e.g., Natural Sciences and Engineering Research Council (NSERC) Discovery Grant program). Often there is *peer review* at this stage, where the proposal and information about the proponent are conveyed to other experts in the field, and an evaluation is provided to the funding agency about (i) the *quality of the proposal*—is it a good question, is it doable, likelihood of success? and (ii) the *quality of the proponent*—what have they done before, have they successfully completed other projects, record of successful graduate student supervision, publication of results in high-quality peer-reviewed journals. Funding programs may specify the relative weighting given to quality of the proposal and proponent, but I think it is often roughly 50:50, with the primary measures of proponent quality being research funding obtained, graduate students supervised, and publications produced.
4. Once the funding is approved for the project, one has to *find the right student or other personnel*. This is not easy, and timing is critical. Graduate studies is the ultimate “learning by doing,” and the student must be technically qualified and committed to the project, as well as able to work with the professor/supervisor and perhaps a partner/client.
5. *Do the project*—easy to say, but often there are many pitfalls, since it is research, requiring solving problems, figuring out new ways, and overcoming vagaries when nature does not cooperate.
6. *Convey the results*. This may mean keeping a partner/client “in the loop” throughout the research process, writing the graduate student thesis, and publishing results of the study in a high-quality peer-reviewed journal. The peer review process here is typically anonymous—a manuscript is submitted to a journal, conveyed to experts (usually two or three) who review the work and recommend acceptance, revision, or rejection of the paper to the journal Editor, who makes the decision. In forestry and

science disciplines, graduate students and professors typically co-author publications. So completing this step augments the professor’s (proponent in step 2 above) record, regarding student successful completion and publications.

This cycle may generate more questions/ideas, which can start the process again. And it is not just a single cycle; active university researchers may have 10 or more of these going on, at different stages, simultaneously.

### Discussion and Lessons Learned

So what are the “levers” in this system, and lessons to be learned? Again, these are my opinions about how the system works.

#### Proponent record matters as much as the proposal

Assuming that a good project idea, a need from either extending scientific understanding or a specific client agency or call for proposals, and a well thought-out research design are all in place, the other equally important piece of the puzzle is the researcher’s record. Allowances are made for young researchers breaking into the system, at least for NSERC Discovery grants in the first two years for young professors. But even here publications from graduate work are expected. And professors who do not take a project to the final publication stage in a timely fashion will have problems with some funding systems.

#### Is there an overemphasis on publications? Quality vs. quantity?

Some think that there is an overemphasis on publishing in academia—“publish or perish.” However, in one sense it is the final external verification of quality of the project. Graduate thesis evaluation also does this, but is less “arms length” and perhaps less rigorous. And in another sense, publication is the only means to ensure wide dissemination of the project and results for posterity. Although theses are disseminated and accessible to a certain degree, many are buried as a single copy in a single university library.

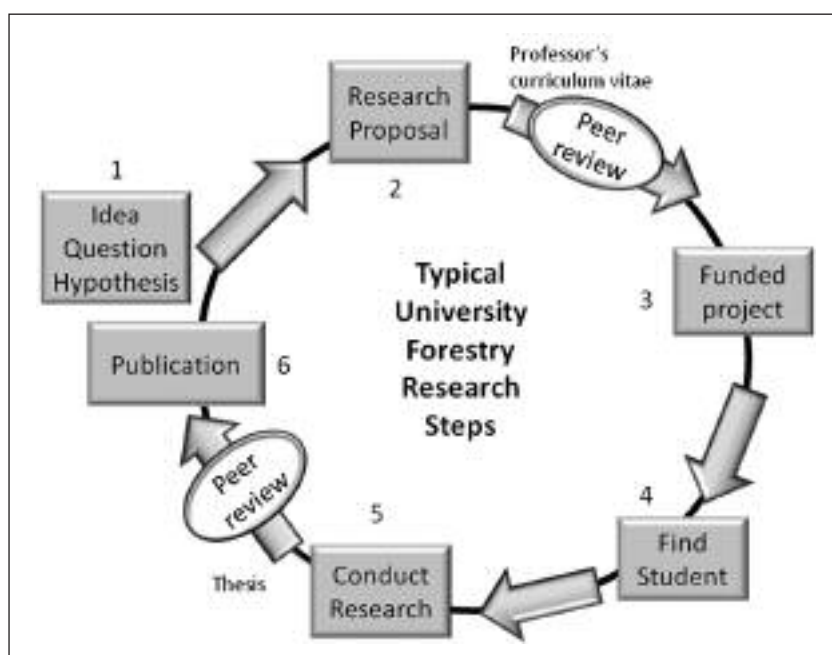


Fig. 1. The research process depicted as a general cycle, with six steps.

Publication quality definitely does count, and all journals and papers are not equal.

Prestige of journals is dependent upon individual forestry disciplines, but in many, papers published in the journals *Nature* and *Science* are the pinnacle for leading-edge research (and in my view, equivalent to perhaps 10 to 20 regular journal papers). Journals such as *Canadian Journal of Forest Research*, *Forest Science*, and *Forest Ecology and Management* are well regarded and suitable for a wide variety of disciplines, and many specialist disciplinary journals are well recognized. And journals like *The Forestry Chronicle* and *Northern Journal of Applied Forestry* may be best for disseminating results to forest practitioners. It is not one size fits all.

#### **Publications are necessary but not sufficient**

In my view, publications are the scientific culmination of the project and set the stage for further work. However, if the target audience is forest industry or government, peer-reviewed publications probably do not fulfill the results dissemination need—they likely have too much detail about “how” and “what was found.” In my experience, such clients want a brief one- or two-page summary emphasizing *what was the question or problem, what was found, and what does it mean?* My students and I have worked extensively on joint projects with J.D. Irving, Ltd. over the last decade, and they like to produce about 500-word, two-panel posters on research projects (see Fig. 2) that are written for a lay audience and posted in Irving District offices and included in an annual “*Irving Forest Research: Discovery Through Science*” supplement to newspapers in New Brunswick.

#### **Why is there research on such esoteric topics?**

Academia values discovery for its own sake, and there are many life-changing research discoveries that began with seemingly esoteric and apparently “useless” discoveries. So agencies such as NSERC support basic research in disciplines that seem somewhat removed from day-to-day life, as well as supporting applied needs-driven research. I think that this mix is a good thing.

#### **Joint projects with industry and government clients**

These are not everybody’s “cup of tea,” but to me personally they are the most rewarding. It is possible to produce both publishable scientific results and make a real-world difference to forest operations and policies. This requires commitment, effort, and sensitivity on both sides (researchers and partners), and in my experience is invariably a two-way learning process, where the researchers learn from partners as well as vice versa. Projects conducted together with J.D. Irving, Ltd. have been designed from the outset with input from their Scientific Advisory Committee, and graduate students regularly present project progress to the Committee and receive feedback from company managers. The company has also made changes to staff job duties, to ensure that a contact is responsible for, and given time for, liaison with researchers. This requires commitment on both sides.

My perception is that industry needs a “comfort level” that it is apt to get useful research results from such collaboration—after all, it involves a significant financial and time commitment—which can make it difficult for researchers to “break in” to collaboration. However, there are some excellent

funding programs based upon researcher/industry collaborative projects, including NSERC’s Collaborative Research and Development and Industrial Post-graduate Scholarship programs that double or triple industry-provided funds (which in themselves are often eligible for substantial tax credits), as well as the Sustainable Forest Management Network Center of Excellence, which requires partner/researcher collaborative projects.

#### **Don’t the rich get richer?**

Yes. It may not be entirely fair, but success is an indicator of future success. Big universities that have a lot of resources typically get more resources. Big provinces that have rich provincial research funding programs or readily available matching funds for national programs that require matching funds help their professors get richer. But all professors have the ability to ensure that their projects complete the cycle in Fig. 1 through to publication, which will increase their future chances. And encouraging graduate students to prepare their research for publication and other dissemination will increase their job prospects as well as bolstering the professor’s résumés.

#### **Research ranking evaluation process at UNB**

Universities value (i) student learning and good teaching; (ii) research, scholarly, and creative activity; and (iii) academic service inside and outside the university. Tenure and promotion assessment processes evaluate all three components, typically for a wide range of possible indicators—it is not one way only, at least at UNB (i.e., we value a mix of contributions).

UNB also has a bi-annual “research ranking” evaluation process for Faculties or Departments, which tracks and tabulates summary data in 22 categories for the current and previous two-year period, along with number of faculty involved. Example categories include: numbers of graduate students, refereed papers, books published, and presentations; amount of research funding; and a variety of other indicators such as adjudication of research grants, papers refereed, organization or chairing of conferences, executive or advisory board positions, etc. A university-wide committee examines data for each category, with respect to: change (positive, negative) from previous period’s data, participation rate within the Department or Faculty, and consistency with other similar units (e.g., comparisons within sciences or within humanities, for units of similar size). The research ranking scale ranges from 1 (highest) to 10, with the extremes described as follows:

(1) *All signs point to major commitment to research and scholarship, indicated through favourable external recognition, an excellent record of adjudicated publications and presented papers, a high level of financial support in those disciplines where such support is appropriate and available, recognition of scholarly expertise through appointments to editorships, grant selection committees, and as journal referees. The graduate program is considered to be of high quality, productive, and attractive to students.*

Versus

(10) *There is a little evidence of research or scholarship on the part of the department as a whole, although there may be isolated cases of good scholarship work by one or more members.*

Research ranking results are linked to Faculty budgets and are also a form of recognition within the university. The





Faculty of Forestry and Environmental Management at UNB first received a research ranking of “1” in 2003–05, and maintained this in the latest 2005–07 period. This is something that most faculty members take pride in, so is one more “lever” in the academic research system.

## Conclusions

This special issue of *The Forestry Chronicle* in commemoration of the Centennial of forestry education at UNB points out some of the changes that have occurred over the years. One major change is the importance attached to research and graduate studies. The first UNB forestry postgraduate degree was conferred on P.Z. Caverhill (BScF 1910) in 1917. Until

1946, a postgraduate degree was conferred upon a candidate who had completed three to five years experience and had submitted a thesis on a forestry subject. Today, all new professors are expected to develop a research program and supervise graduate students. Professors are also expected to successfully raise the research funds to support those students, typically at about \$18 000 to \$21 000 per year student support plus research support. Funding limits current forestry graduate student enrolment at UNB, but was still about \$4.5 million per year over the last two years. Hopefully, comments in this article will help graduate students and professors with research success, and help practitioners understand what makes the “forestry research game” run.